

MD MERAZ AHASAN SHAH

Software Engineer

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[Portfolio](#) | [GitHub](#) | [LinkedIn](#)

Professional Summary

Computer Science and Engineering undergraduate with hands-on experience in software engineering, backend development, and computer vision research. Skilled in developing secure and database-driven applications using React, Next.js, NestJS, Express.js, PostgreSQL, and MongoDB. Experienced in RESTful API design, JWT authentication, relational database modeling, and deep-learning-based object detection.

Technical Skills

Languages	JavaScript, TypeScript, Python, C#, C++, Java, SQL
Frameworks	React, Next.js, NestJS, Express.js, ASP.NET Core, Tailwind CSS, Vite
Backend	Node.js, RESTful APIs, JWT Authentication, API Integration
Databases	PostgreSQL, MongoDB, MySQL, Microsoft SQL Server, Firebase
AI / ML	PyTorch, Ultralytics YOLO, Roboflow, Computer Vision, Object Detection
Tools	Git, GitHub, Docker, Postman, VS Code, npm
Core	OOP, Data Structures, Algorithms, MVC, Clean Architecture, CRUD

Selected Projects

Job Portal System	NestJS, React, PostgreSQL
GitHub	

- Engineered a recruitment platform enabling employers to publish jobs and candidates to search and apply for opportunities.
- Developed RESTful APIs with JWT authentication, role-based authorization, input validation, and PostgreSQL data models.

Codveda Web Development	React, Node.js, Express.js, MongoDB
GitHub	

- Completed six web development projects covering responsive interfaces, JavaScript applications, React components, and backend services.
- Implemented CRUD workflows and JWT-based authentication with protected routes and password hashing.

LifeDrop	Kotlin, Firebase
GitHub	

- Developed an Android blood donation platform connecting donors with users requiring emergency blood support.
- Integrated Firebase Authentication and cloud storage for secure registration, account management, and user data.

Research Thesis

BDFD-10: A Bangladesh-Centric Multi-Class Fabric Defect Dataset and a Cross-Generation Benchmark of Deep Object Detectors	2025–2026
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- Developed BDFD-10, a Bangladesh-centric fabric defect dataset containing **1,208 real-world images** collected from four textile and garment factory groups.
- Benchmarked **10 object detection models**, including YOLOv8, YOLO11, YOLO12, YOLO26, RT-DETR-L, and Faster R-CNN under a unified evaluation protocol.
- Achieved **93.2% mAP@0.5** using YOLOv8s and implemented the data preparation and evaluation pipeline with Python, PyTorch, Ultralytics YOLO, Roboflow, and Google Colab.

Education

American International University-Bangladesh	2022–Present
B.Sc. in Computer Science and Engineering	Expected Graduation: 2026
Cantonment Public School and College	2019
Higher Secondary Certificate (HSC)	GPA: 4.67 / 5.00
Lions School and College	2017
Secondary School Certificate (SSC)	GPA: 5.00 / 5.00

Achievements

30+ Academic and Personal Projects | 250+ GitHub Contributions | Bachelor's Research in Computer Vision

Languages

Bengali: Native **English:** Professional Working Proficiency